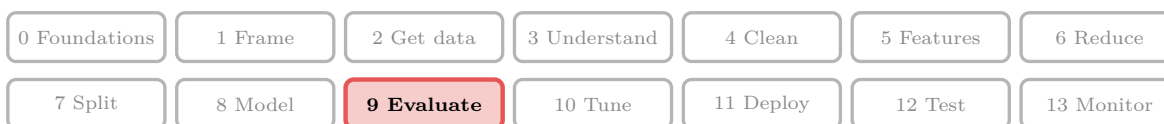


# Classification Metrics: Accuracy and the Confusion Matrix

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**Where this fits:** Pipeline map step 9 (Evaluate). See the [Course map](#).



- **Builds on:** Imbalanced data ([Note 9](#)); Train-test split ([Note 13](#)).
- **Leads to:** Precision, recall and F1 ([Note 77](#)); ROC curve and AUC ([Note 78](#)); ANN for classification ([Note 1011](#)).
- **Compare with:** Type I and II errors, power, tails ([Note 292](#)).

## 1. Overview

**Key point:** Accuracy is the fraction of correct predictions. The confusion matrix splits the predictions into four counts, which also show what kind of mistakes a model makes.

Regression models were judged with regression metrics such as  $R^2$  (the regression metrics Note). Classification models need their own **classification metrics**. There are several: accuracy, the confusion matrix, precision, recall, F1 score, ROC-AUC and more.

This Note covers the first two. The following Notes build on the confusion matrix.

## 2. Accuracy

**Key point:** Accuracy = number of correct predictions / total number of predictions.

### 2.1 The idea

**Key point:** Compare each prediction with the true label and count how many match.

Suppose we train two models on the student placement data, for example logistic regression and a decision tree (a later Note), and predict the test students. For each test student we know the true result, so we can tick each prediction as right or wrong.

$$\text{accuracy} = \frac{\text{number of correct predictions}}{\text{total number of predictions}}$$

With numbers: if a model gets 8 of 10 test students right, its accuracy is  $8/10 = 0.8$ , or 80%. A model with 9 of 10 right (90%) is better on this test set.

### 2.2 A real example

**Key point:** On the heart-disease data, logistic regression reaches 0.869 and a decision tree 0.836.

The heart-disease dataset has 303 patients, 13 medical measurements each (age, blood pressure, cholesterol and so on), and a target: 1 if the patient has heart disease, 0 if not. We split off 20% as the test set (random state 2) and train both models.

**Python:** Accuracy in scikit-learn.

```
from sklearn.metrics import accuracy_score

lr = make_pipeline(StandardScaler(), LogisticRegression())
lr.fit(X_train, y_train)
accuracy_score(y_test, lr.predict(X_test))    # 0.869
```

| Model                                     | Test accuracy |
|-------------------------------------------|---------------|
| Logistic regression (inputs standardised) | 0.869         |
| Decision tree                             | 0.836         |

So, on this test set, logistic regression is the better model.

## 2.3 More than two classes

**Key point:** Accuracy is computed the same way for any number of classes.

Nothing changes with more classes, for example the three iris species: count the correct predictions and divide by the total. Logistic regression on the iris test set (30 flowers) gets 29 right: accuracy 0.967.

## 3. How much accuracy is good enough?

**Key point:** It depends on the problem. 99% can be far too low for cancer detection and plenty for predicting food orders.

A common interview question is “what accuracy should a model have?”. There is no single number; it depends on the cost of a mistake.

- **Cancer detection from X-rays:** 99% accuracy means one patient in a hundred gets a wrong answer. That is not acceptable.
- **Self-driving car decisions:** 99% means an error every hundred decisions. Also not acceptable.
- **Predicting whether a customer orders food this weekend:** 80% may be perfectly useful, because a wrong guess costs little.

## 4. The confusion matrix

**Key point:** A table of actual classes against predicted classes. For two classes it has four counts: TN, FP, FN and TP.

### 4.1 What accuracy hides

**Key point:** Accuracy says how many mistakes there are, not which kind.

A model with 90% accuracy makes mistakes 10% of the time. But in a two-class problem there are two kinds of mistake:

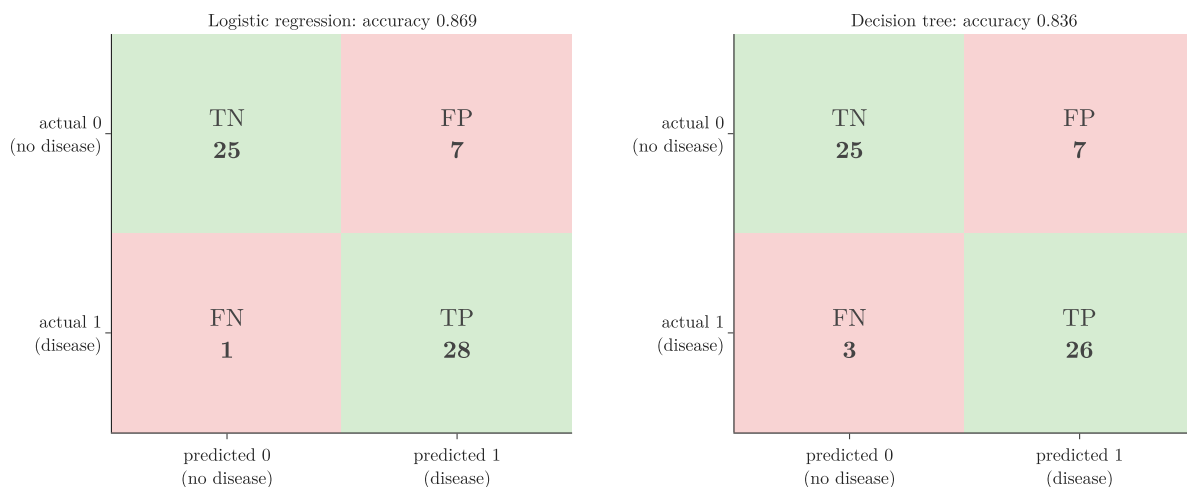
- a patient **has** heart disease, but the model says no;
- a patient does **not** have heart disease, but the model says yes.

These have very different consequences, and accuracy cannot tell them apart. The **confusion matrix** can.

### 4.2 Reading it

**Key point:** Rows are the actual classes, columns the predicted ones. The diagonal holds the correct predictions.

Figure 1 shows the confusion matrices of both models on the heart-disease test set (61 patients).



**Figure 1:** Confusion matrices for logistic regression and the decision tree

In scikit-learn's layout, each **row** is an actual class and each **column** a predicted class. For logistic regression:

|                       | Predicted 0 (no disease) | Predicted 1 (disease)   |
|-----------------------|--------------------------|-------------------------|
| Actual 0 (no disease) | 25: true negatives (TN)  | 7: false positives (FP) |
| Actual 1 (disease)    | 1: false negatives (FN)  | 28: true positives (TP) |

- **True positives (28):** patients with heart disease that the model correctly flagged.
- **True negatives (25):** healthy patients correctly cleared.
- **False positives (7):** healthy patients wrongly flagged as ill.
- **False negatives (1):** an ill patient the model missed.

The decision tree makes the same 7 false positives but 3 false negatives. So most of its extra errors are of the more dangerous kind: missed patients.

**Extra:** Some books and websites draw the matrix the other way round, with predictions as rows. Always check the axis labels before reading one.

### 4.3 The names

**Key point:** The second word is what the model predicted; the first word says whether that was right.

True / False      Positive / Negative

first word: was the  
prediction **right**?

second word: what  
the model **predicted**

False Positive: predicted 1, wrong (really 0) = Type I error  
False Negative: predicted 0, wrong (really 1) = Type II error

**Figure 2:** How the four names are built

- **Positive / Negative** is the model's prediction: 1 or 0.
- **True / False** says whether that prediction was correct.

So a “false positive” is a prediction of 1 that was wrong (the truth was 0). A “true negative” is a prediction of 0 that was right.

### 4.4 Type I and Type II errors

**Key point:** A false positive is a Type I error; a false negative is a Type II error.

These two error types also have names from statistics:

| Error                                 | Meaning              | Heart example                                  |
|---------------------------------------|----------------------|------------------------------------------------|
| <b>Type I error</b> = false positive  | predicted 1, truly 0 | a healthy patient told they have heart disease |
| <b>Type II error</b> = false negative | predicted 0, truly 1 | an ill patient told they are healthy           |

#### 4.5 Accuracy from the confusion matrix

**Key point:** Accuracy =  $(TP + TN) / (TP + TN + FP + FN)$ : the diagonal over the total.

The correct predictions are on the diagonal, so

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{28 + 25}{28 + 25 + 7 + 1} = \frac{53}{61} = 0.869$$

The confusion matrix gives the accuracy, but the accuracy cannot give back the confusion matrix: it carries less information.

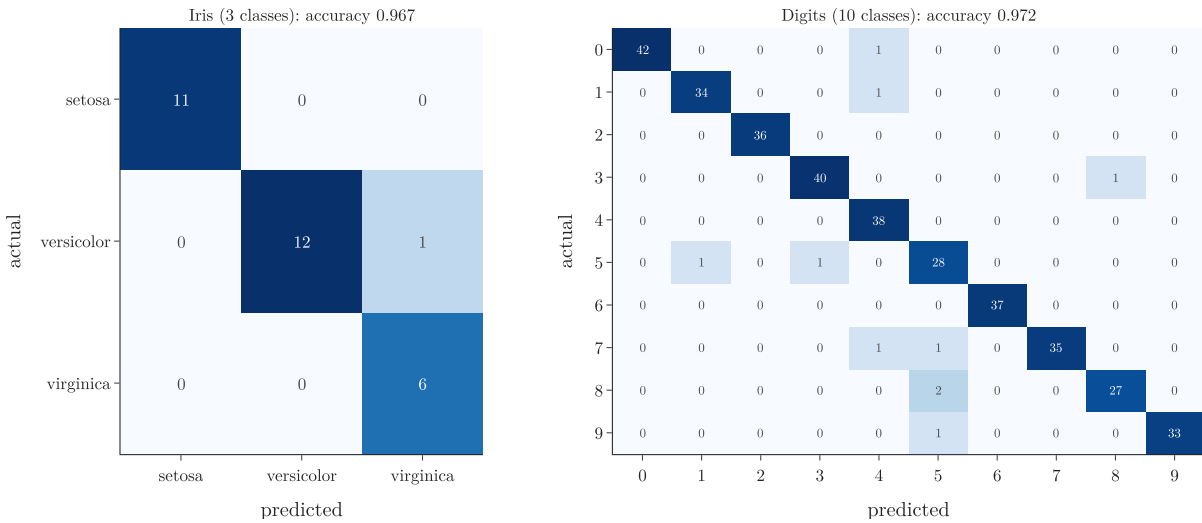
**Python:** The confusion matrix in scikit-learn.

```
from sklearn.metrics import confusion_matrix

confusion_matrix(y_test, lr.predict(X_test))
# [[25  7]
#   [ 1 28]]
tn, fp, fn, tp = confusion_matrix(y_test, y_pred).ravel()
```

#### 5. More than two classes

**Key point:** With k classes the confusion matrix is  $k \times k$ . The diagonal is still the correct predictions; every other cell is a specific confusion.



**Figure 3:** Confusion matrices for iris (3 classes) and digits (10 classes)

- **Iris** (left):  $3 \times 3$ . The one mistake is a versicolor flower predicted as virginica (row versicolor, column virginica).
- **Digits** (right):  $10 \times 10$ , from scikit-learn’s  $8 \times 8$  pixel images of handwritten digits. Logistic regression gets 97.2% right. The off-diagonal cells show exactly which digits get confused: for example, two 8s were read as 5s.

Accuracy is again the diagonal total divided by the grand total.

## 6. When accuracy misleads: imbalanced data

**Key point:** If one class is very rare, a model that always predicts the common class gets very high accuracy while being useless.

Data is **imbalanced** when one class is much rarer than the other.

Imagine a model that screens airline passengers for security threats. Out of 100,000 passengers, perhaps 10 are real threats. A “model” that simply says “not a threat” for everyone gets

|          | Predicted 0 | Predicted 1 |
|----------|-------------|-------------|
| Actual 0 | 99,990      | 0           |
| Actual 1 | 10          | 0           |

$$\text{accuracy} = \frac{99,990}{100,000} = 0.9999$$

An accuracy of 99.99%, for a model that catches none of the 10 threats. The confusion matrix shows the problem at once: all 10 positives are false negatives.

So on imbalanced data, accuracy alone is the wrong metric. The next Note introduces **precision** and **recall**, which focus on the rare class.

## 7. Summary

| Term         | Meaning                    | Heart, logistic regression |
|--------------|----------------------------|----------------------------|
| TP           | predicted 1, truly 1       | 28                         |
| TN           | predicted 0, truly 0       | 25                         |
| FP (Type I)  | predicted 1, truly 0       | 7                          |
| FN (Type II) | predicted 0, truly 1       | 1                          |
| Accuracy     | $(TP + TN) / \text{total}$ | 0.869                      |

- Accuracy is simple and works for any number of classes, but “good enough” depends on the problem.
- The confusion matrix shows the kind of each mistake; rows are actual, columns predicted in scikit-learn.
- On imbalanced data, accuracy can be near 100% for a useless model.

## 8. Key terms

| Term                  | Meaning                                                                   |
|-----------------------|---------------------------------------------------------------------------|
| Classification metric | A number that measures how well a classification model performs           |
| Accuracy              | The fraction of predictions that are correct                              |
| Confusion matrix      | A table counting predictions for every pair of actual and predicted class |
| True positive (TP)    | Predicted positive, and actually positive                                 |
| True negative (TN)    | Predicted negative, and actually negative                                 |
| False positive (FP)   | Predicted positive, but actually negative; a Type I error                 |
| False negative (FN)   | Predicted negative, but actually positive; a Type II error                |
| Imbalanced data       | Data in which one class is much rarer than another                        |